Annual Facility Report 2024

Lakeside Research Reactor · Published March 2025

Summary

The reactor operated for 214 days in 2024, compared with 251 days in 2023. Operation supported 38 research projects and the production of reference samples for three partner laboratories.

There were no reportable safety events. Two minor equipment faults were corrected during scheduled maintenance.

1. Operations

The planned maintenance outage began in May and was extended by six weeks to complete replacement of the secondary heat exchanger. The extension accounts for most of the reduction in operating days.

Fuel inventory, core configuration, and experimental facilities were unchanged during the year.

2. Occupational exposure

Average annual dose for monitored workers rose from 0.42 mSv in 2023 to 0.61 mSv in 2024. The highest individual dose was 2.3 mSv.

Year	Average dose	Highest dose	Monitored workers
2022	0.39 mSv	1.8 mSv	61
2023	0.42 mSv	1.9 mSv	64
2024	0.61 mSv	2.3 mSv	66

All individual doses remained below the facility administrative limit of 5 mSv per year, which is set well below the regulatory limit.

Collective dose figures are reported in the appendix.

3. Training and staffing

Twelve new staff completed initial radiation safety training. Refresher completion reached 97 percent by year end; the remaining staff completed refreshers in January 2025.

4. Waste and effluents

Solid waste volumes were similar to 2023. Gaseous and liquid effluents remained below one percent of authorised discharge limits.

5. Looking ahead

Priorities for 2025 include completing the instrumentation upgrade, revising the emergency preparedness plan, and reviewing the dosimetry service contract.

This report will be updated if the appendix figures change after final verification.